CE725 Construction Economics and Finance: Project Team Charter

Course Project, Annexure B format. Original charter submitted at topic selection (rows 1 to 5); rows 6 to 10 completed 5 November 2026 to reflect the tool as built.

Item	Details
Team number	8
Team members and roles	Maneesha (23B4223), R1 Project Lead and Finance Modeller. Dinesh Sai (23B1022), R2 Data Lead. Sujay (23B1059), R3 AI and Tool Developer. Surya Theja (23B1062), R4 Validation Engineer. Akshitha (23B0629), R5 Documentation and Reflection Lead.
Topic selected	Topic 2: Real Estate Project Feasibility and Financial Appraisal Tool
Problem statement (100 words)	Real estate developers commit substantial capital before knowing how land cost, construction cost, approvals, financing, taxes, selling prices, sales absorption and project timelines will unfold. A single headline figure such as IRR can make a project look viable while hiding how fragile that result is. Our project develops an AI-assisted feasibility and decision-support tool that integrates these inputs into a transparent cash flow model, calculating NPV, IRR, payback period, profit on cost and break-even selling price. It tests robustness through scenario, sensitivity and stress analysis, identifies the key risk drivers, and generates a plain-language investment memo to support developer decisions.
Intended users of the tool	Primary: real estate developers and promoters deciding whether to commit to land and approvals. Secondary: lenders and investors assessing viability and downside resilience; feasibility analysts and consultants; and non-technical decision-makers (directors, partners) who receive the plain-language investment memo.
Method(s) to be applied	Monthly discounted cash flow at the equity level over the project life, with an unlevered project view alongside. Metrics: NPV at a hurdle rate, IRR with detection of non-unique IRR, payback and discounted payback, profit on cost, and break-even selling price by bisection. One-at-a-time sensitivity on eight drivers, a tornado chart, three preset stress scenarios and a custom scenario. A plain-language investment memo generated from a fact sheet of the computed numbers, with every number in the memo checked back against the model.
Data sources	Published figures for a peripheral Mumbai Metropolitan Region project, checked on 5 November 2026 and listed with links in `docs/DATA_SOURCES.md`: Knight Frank India Real Estate H2 2025 (prices, absorption); JLL 2024 land-transaction data (land per acre); JLL Construction Cost Guide 2026 and cost-calculator sites (construction cost); MCHI-CREDAI approval-cost paper (approvals); RBI monetary policy statement of 7 October 2026 and lender rate cards (interest); Income Tax Act s.115BAA (tax); EY/NSE Cost of Capital Survey 2024 (discount rate); RERA Act 2016 s.13 and MahaRERA payment schedules (collections).
Build platform	Python 3.12 with Streamlit for the interface, pandas and numpy for the engine, Plotly for charts, openpyxl for the benchmark workbook, pytest and Playwright for tests. Hosted on Streamlit Community Cloud (https://ce725-real-estate-feasibility.streamlit.app/) with source on GitHub. Optional Claude memo through the AIML API. Excel or LibreOffice Calc for the benchmark workbook.
Benchmark case for validation	1,00,000 sq ft mid-segment project: land ₹20 cr in month 1; approvals ₹3 cr over months 1 to 3; construction ₹3,000/sq ft over months 4 to 30 on an S-curve; sales at ₹7,500/sq ft from month 7 to 30 with 20% at booking and the balance over 6 months; 2% marketing; 50% debt at 12% with a 50% collection sweep; 25% tax; 15% discount rate. Validated by an Excel workbook of pure formulas, three further input cases through the same formulas, an exact-arithmetic recomputation, and a hand calculation of months 1, 8 and 36 and the NPV by the Validation Engineer.
Key risks to completion	Unverified AI output entering the model or reports (mitigated by the benchmark layers and the number check on the memo). IRR being non-unique for leveraged cash flows (detected and flagged by the tool). Base-case inputs not matching a real market (mitigated by the data source list, with three optimistic inputs declared). Time, with the reflection report the last deliverable to finish. Dependence on an external API for the memo (the template memo works offline).